GCSE Mathematics practice paper

Answer all questions. Show your working.

Question 1 (2 marks)

Simplify $x^5 \times x^3$.

Answer: x^8

Question 2 (3 marks)

Solve the simultaneous equations $2x + y = 11$ and $x - y = 1$.

Answer: $x = 4, y = 3$

AI-generated worked solutions - may contain mistakes

Question 3 (2 marks)

Work out $\frac{3}{4} + \frac{1}{8}$. Give your answer as a fraction.

Answer: $7/8$

Question 4 (3 marks)

A coat costs £80. In a sale the price is reduced by 15%. Work out the sale price.

Answer: 68

(Not checked by calculation: look over this one)

Question 5 (2 marks)

Write 0.000 52 in standard form.

Answer: 5.2×10^{-4}

AI-generated worked solutions

Written by an AI and may contain mistakes. Answers marked as not confidently solved or not checked need a careful look.

Question 1 (2 marks)

Simplify x^5 × x^3.

Checked by calculation

When multiplying powers with the same base, add the exponents.

$x^5 \times x^3 = x^{5+3}$

Calculate the sum of the exponents.

$5 + 3 = 8$

Answer: x^8

To simplify the expression, add the indices (exponents) of the like bases together.

Question 2 (3 marks)

Solve the simultaneous equations 2x + y = 11 and x - y = 1.

Checked by calculation

Add the two equations together to eliminate y.

$(2x + y) + (x - y) = 11 + 1$

Simplify and solve for x.

3x = 12 \implies x = 4

Substitute x = 4 into the second equation.

$4 - y = 1$

Rearrange to find y.

-y = -3 \implies y = 3

Answer: $x = 4, y = 3$

Add the equations to eliminate y, solve for x, then substitute back to find y.

Question 3 (2 marks)

Question 3 (2 marks) Work out 3/4 + 1/8. Give your answer as a fraction.

Checked by calculation (against the AI's reading of the question)

Find the common denominator for 4 and 8, which is 8.

$3/4 = (3 * 2)/(4 * 2) = 6/8$

Now add the two fractions together.

$6/8 + 1/8 = 7/8$

Answer: $7/8$

To add fractions, find a common denominator and then add the numerators.

Question 4 (3 marks)

Question 4 (3 marks)\nA coat costs £80. In a sale the price is reduced by 15%. Work out the sale price.

Not checked by calculation: look over this one

Find 15% of 80.

$15\% \text{ of } 80 = \frac{15}{100} \times 80 = 12$

Subtract this amount from the original price.

$80 - 12 = 68$

Answer: 68

To find the sale price, calculate 15% of £80 and subtract it from the original cost.

Question 5 (2 marks)

Write 0.000 52 in standard form.

Checked by calculation

Move the decimal point to the right until there is one non-zero digit before it.

$0.00052 \rightarrow 5.2$

Count the number of places moved, which is 4.

$The decimal moved 4 places.$

Since we moved right, the exponent is negative.

5.2×10^{-4}

Answer: 5.2×10^{-4}

To write a number in standard form, move the decimal point so that there is one non-zero digit to its left, then multiply by 10 raised to the power of the number of places moved (negative if moved right).